Program Source Code:

```
#include <stdio.h>

int main() {
    int id, pin, tries = 0, choice, receiver;
    int accountNo = 1001;
    float balance = 5000, amount;

    while (tries < 3) {
        printf("Enter User ID: ");
        scanf("%d", &id);
        printf("Enter PIN: ");
        scanf("%d", &pin);

        if (id == 101 && pin == 1234)
            break;

        printf("Invalid login! Try again.\n");
        tries++;
    }

    if (tries == 3) {
        printf("Access denied.\n");
        return 0;
    }

    do {
        printf("\n--- BANKING MENU ---\n");
        printf("1. Check Balance\n2. Deposit Money\n");
        printf("3. Withdraw Money\n4. Transfer Money\n");
        printf("5. Account Information\n6. Exit\n");
        printf("Enter choice: ");
        scanf("%d", &choice);

        switch (choice) {
            case 1:
                printf("Balance: Rs. %.2f\n", balance);
                break;

            case 2:
                printf("Enter deposit amount: ");
                scanf("%f", &amount);
                if (amount > 0) {
                    balance += amount;
                    printf("Deposit successful.\n");
                } else {
                    printf("Invalid amount.\n");
                }
                break;

            case 3:
                printf("Enter withdrawal amount: ");
                scanf("%f", &amount);
                if (amount <= 0)
                    printf("Invalid amount.\n");
                else if (amount > balance)
                    printf("Insufficient balance.\n");
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                else {
                    balance -= amount;
                    printf("Withdrawal successful.\n");
                }
                break;

            case 4:
                printf("Enter receiver account number: ");
                scanf("%d", &receiver);
                printf("Enter transfer amount: ");
                scanf("%f", &amount);
                if (receiver <= 0 || receiver == accountNo)
                    printf("Invalid receiver account.\n");
                else if (amount <= 0)
                    printf("Invalid amount.\n");
                else if (amount > balance)
                    printf("Insufficient balance.\n");
                else {
                    balance -= amount;
                    printf("Transfer successful to account %d.\n",
                           receiver);
                }
                break;

            case 5:
                printf("Account Number: %d\n", accountNo);
                printf("Account Status: Active\n");
                printf("Current Balance: Rs. %.2f\n", balance);
                break;

            case 6:
                printf("Thank you for using the bank system.\n");
                break;

            default:
                printf("Invalid choice. Please try again.\n");
        }
    } while (choice != 6);

    return 0;
}
```

Program Source Code:

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>


#define MAX_ACCOUNTS 5
#define NAME_LENGTH 50
#define FILE_NAME "bank_accounts.dat"


typedef struct {
    int accountNo;
    char name[NAME_LENGTH];
    int pin;
    double balance;
} Account;


Account accounts[MAX_ACCOUNTS] = {
    {1001, "Aarav Sharma", 1234, 5000.00},
    {1002, "Priya Verma", 2345, 7500.00},
    {1003, "Rahul Patel", 3456, 6200.00}
};
int accountCount = 3;


/* Function prototypes */
int login(int *loggedInIndex);
void showMenu(void);
void checkBalance(const Account *account);
void deposit(Account *account);
void withdraw(Account *account);
void transfer(Account *sender);
void showAccount(const Account *account);
void showAllAccounts(void);
void saveAccounts(void);
void loadAccounts(void);
int findAccount(int accountNo);
void printLine(void);


int main(void) {
    int loggedInIndex = -1;
    int choice;


    loadAccounts();


    printf("\n====================================\n");
    printf("       SIMPLE BANKING SYSTEM V2\n");
    printf("====================================\n");


    if (!login(&loggedInIndex)) {
        printf("\nAccess denied. Too many incorrect attempts.\n");
        return 0;
    }


    do {
        showMenu();
        printf("Enter your choice: ");


        if (scanf("%d", &choice) != 1) {
            printf("Invalid input. Please enter a number.\n");
            int ch;
            while ((ch = getchar()) != '\n' && ch != EOF) { }
            choice = 0;
            continue;
        }


        switch (choice) {
            case 1:
                checkBalance(&accounts[loggedInIndex]);
                break;
            case 2:
                deposit(&accounts[loggedInIndex]);
                break;
            case 3:
                withdraw(&accounts[loggedInIndex]);
                break;
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            case 4:
                transfer(&accounts[loggedInIndex]);
                break;
            case 5:
                showAccount(&accounts[loggedInIndex]);
                break;
            case 6:
                showAllAccounts();
                break;
            case 7:
                saveAccounts();
                printf("Data saved. Thank you for using the system.\n");
                break;
            default:
                printf("Invalid choice. Select 1 to 7.\n");
        }
    } while (choice != 7);

    return 0;
}

int login(int *loggedInIndex) {
    int enteredId, enteredPin, attempts;

    for (attempts = 1; attempts <= 3; attempts++) {
        printf("\nEnter account number: ");
        if (scanf("%d", &enteredId) != 1) {
            printf("Invalid account number format.\n");
            int ch;
            while ((ch = getchar()) != '\n' && ch != EOF) { }
            continue;
        }

        printf("Enter PIN: ");
        if (scanf("%d", &enteredPin) != 1) {
            printf("Invalid PIN format.\n");
            int ch;
            while ((ch = getchar()) != '\n' && ch != EOF) { }
            continue;
        }

        int index = findAccount(enteredId);
        if (index != -1 && accounts[index].pin == enteredPin) {
            *loggedInIndex = index;
            printf("\nLogin successful. Welcome, %s!\n", accounts[index].name);
            return 1;
        }
        printf("Incorrect account number or PIN. Attempt %d of 3.\n", attempts);
    }
    return 0;
}

void showMenu(void) {
    printLine();
    printf("\n1. Check Balance\n");
    printf("2. Deposit Money\n");
    printf("3. Withdraw Money\n");
    printf("4. Transfer Money\n");
    printf("5. View My Account\n");
    printf("6. View All Accounts (Demo)\n");
    printf("7. Save and Exit\n");
    printLine();
}

void checkBalance(const Account *account) {
    printf("\nCurrent balance: Rs. %.2f\n", account->balance);
}

void deposit(Account *account) {
    double amount;
    printf("Enter deposit amount: Rs. ");
    if (scanf("%lf", &amount) != 1) {
        printf("Invalid amount format.\n");
        int ch;
        while ((ch = getchar()) != '\n' && ch != EOF) { }
        return;
    }
    if (amount <= 0) {
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        printf("Amount must be greater than zero.\n");
        return;
    }
    account->balance += amount;
    printf("Deposit successful. New balance: Rs. %.2f\n", account->balance);
}

void withdraw(Account *account) {
    double amount;
    printf("Enter withdrawal amount: Rs. ");
    if (scanf("%lf", &amount) != 1) {
        printf("Invalid amount format.\n");
        int ch;
        while ((ch = getchar()) != '\n' && ch != EOF) { }
        return;
    }
    if (amount <= 0) {
        printf("Amount must be greater than zero.\n");
    } else if (amount > account->balance) {
        printf("Insufficient balance.\n");
    } else {
        account->balance -= amount;
        printf("Withdrawal successful. New balance: Rs. %.2f\n", account->balance);
    }
}

void transfer(Account *sender) {
    int receiverNo, receiverIndex;
    double amount;


    printf("Enter receiver account number: ");
    if (scanf("%d", &receiverNo) != 1) {
        printf("Invalid account number.\n");
        int ch;
        while ((ch = getchar()) != '\n' && ch != EOF) { }
        return;
    }


    receiverIndex = findAccount(receiverNo);
    if (receiverIndex == -1 || receiverNo == sender->accountNo) {
        printf("Receiver account not found or invalid.\n");
        return;
    }


    printf("Enter transfer amount: Rs. ");
    if (scanf("%lf", &amount) != 1) {
        printf("Invalid amount format.\n");
        int ch;
        while ((ch = getchar()) != '\n' && ch != EOF) { }
        return;
    }


    if (amount <= 0) {
        printf("Amount must be greater than zero.\n");
    } else if (amount > sender->balance) {
        printf("Insufficient balance.\n");
    } else {
        sender->balance -= amount;
        accounts[receiverIndex].balance += amount;
        printf("Transfer of Rs. %.2f to account %d completed.\n",
               amount, receiverNo);
        printf("Updated balance: Rs. %.2f\n", sender->balance);
    }
}

void showAccount(const Account *account) {
    printf("\nACCOUNT DETAILS\n");
    printLine();
    printf("Account number : %d\n", account->accountNo);
    printf("Account holder : %s\n", account->name);
    printf("Account status : Active\n");
    printf("Balance        : Rs. %.2f\n", account->balance);
}

void showAllAccounts(void) {
    int i;
    printf("\nDEMO ACCOUNT LIST\n");
    printLine();
```

```
    for (i = 0; i < accountCount; i++) {
        printf("%d | %-20s | Rs. %.2f\n",
               accounts[i].accountNo, accounts[i].name,
               accounts[i].balance);
    }
}


int findAccount(int accountNo) {
    int i;
    for (i = 0; i < accountCount; i++) {
        if (accounts[i].accountNo == accountNo)
            return i;
    }
    return -1;
}


void saveAccounts(void) {
    FILE *fp = fopen(FILE_NAME, "wb");
    if (fp == NULL) {
        printf("Error: unable to open file for saving.\n");
        return;
    }
    fwrite(&accountCount, sizeof(int), 1, fp);
    fwrite(accounts, sizeof(Account), accountCount, fp);
    fclose(fp);
}


void loadAccounts(void) {
    FILE *fp = fopen(FILE_NAME, "rb");
    if (fp == NULL)
        return; /* First run: use the sample accounts above. */


    int storedCount;
    if (fread(&storedCount, sizeof(int), 1, fp) == 1 &&
        storedCount > 0 && storedCount <= MAX_ACCOUNTS) {
        size_t readCount = fread(accounts, sizeof(Account), storedCount, fp);
        if (readCount == (size_t)storedCount)
            accountCount = storedCount;
    }
    fclose(fp);
}


void printLine(void) {
    printf("-----------------------------------\n");
}
```